RAILBLOCK AI — SYSTEM & ARCHITECTURAL REPORT

Intelligent Block Planning & Multi-Departmental Operations for Indian Railways (SIH26027)

Generated: November 2026	**Platform:** RailBlock AI v0.2.0	**Target Division:** Mumbai Suburban (CR/WR/HR)	**Audit Status:** Verified (23/23 Tests Passed)

1. Executive Summary & Purpose

RailBlock AI is an automated operations research and decision-support platform engineered for the Ministry of Railways (SIH Problem Statement **SIH26027**). The system harmonizes maintenance demands across three traditionally siloed railway engineering disciplines: **Engineering (Civil/Track/P-Way)**, **Traction Distribution (TRD/Electrical 25kV OHE)**, and **Signal & Telecommunication (S&T;)**. By pairing mathematical constraint optimization (Google OR-Tools CP-SAT) with realistic timetable detention modeling, the platform automatically schedules maintenance blocks during shadow windows, eliminating corridor gridlocks while ensuring track safety.

This report provides an end-to-end operational scan of how all subsystems function—spanning frontend React interfaces, backend FastAPI endpoints, database schemas, and AI optimization engines—alongside an actionable list of audit findings and prioritized improvements.

2. System Architecture & Component Workflow

The application operates in a decoupled client-server architecture designed for high availability and rapid user interaction:

Subsystem	Core Technologies	Key Role & Mechanism
Frontend UI	React 18, Vite, TypeScript, Tailwind CSS, Recharts	Responsive dual-mode portal (Admin Dispatcher vs Public Passenger). Features customizable color palette (Pantone Wheat, Magnetic Blue, Carrot Orange, and Pure Black night mode), interactive Gantts, and drawer modals.
Backend API	Python 3.12, FastAPI, Uvicorn, Pydantic v2	High-performance asynchronous REST API handling data ingestion, corridor aggregation, scheduling dispatches, and real-time WebSocket telemetry.
Constraint Solver	Google OR-Tools CP-SAT Engine	Formulates block windows as integer interval variables. Solves multi-objective optimization balancing safety urgency, gross tonnage (GMT), and train detention penalties in seconds.
Priority Engine	Deterministic Multi-Criteria Scoring (MCA)	Computes composite priority scores (0–100) based on ultrasonic rail flaws (USFD), track geometry indices, contact wire wear, overdue days, and speed cautions.
Telemetry Engine	Python Asyncio + WebSockets + Kavach Simulator	Streams simulated live train positions, block occupancy, and emergency track flaw broadcasts across Mumbai Central, Western, and Harbour corridors.

Persistence Layer	SQLAlchemy 2.0 + SQLite (Prod: PostgreSQL ready)	Relational models covering Corridors, Physical Track Assets, Maintenance Demands, Timetables, Block Windows, and Master Plans.

3. Functional View Analysis (Frontend Modules)

The platform provides 10 purpose-built views organized into three operational domains:

Domain	View Name	Operational Capabilities & Features
Operations	**Dashboard**	Executive command center with 4 live KPI sparklines, Admin Action Center for 1-click bottleneck mitigation, collapsible real-time Kavach telemetry simulator, 7-day multi-department Gantt, and sector workload distribution.
Operations	**Scheduler**	Interactive OR-Tools control station. Allows tuning safety buffers, solver timeouts, joint maintenance bundling, and running the CP-SAT engine with instant Gantt timeline visualization.
Operations	**Conflicts View**	Automated timetable clash detector. Identifies train-maintenance overlaps, predicts detention minutes, and recommends verified mitigation strategies (reroute, reschedule, or cancel).
Operations	**Blocks & Timetable**	Full timetable inspection matrix for Vande Bharat, Rajdhani, Suburban Locals, and Freight trains across Central, Western, and Harbour lines.
Planning	**Tasks Registry**	Master repository of 240+ maintenance requests with multi-criteria priority scoring, status badges, department filtering, and slide-over task detail inspection drawer.
Planning	**Plans & Compare**	Side-by-side plan comparator contrasting manual baseline scheduling against OR-Tools AI plans. Displays metrics on train delay reduction, block utilization improvement, and asset availability.
Administration	**Sector Assignment**	Field dispatch Kanban board with 5 operational workflow stages (Requisitioned, Assigned, Permit Issued, WIP, Fit Certified). Features bounded scroll containers and Cards/Compact view toggle.
Administration	**Assets View**	Physical asset inventory covering track sections, turnouts, OHE catenary masts, electronic interlocking points, and digital axle counters with degradation indices.
Administration	**Data Pipeline**	Dataset management portal: OpenStreetMap (OSM) track geometry importer, Central/Western press release scraper for Sunday mega blocks, and live database sync monitor.
Passenger	**Public Viewer**	Commuter-facing portal showing live regulated local trains, planned weekend mega blocks, and passenger advisory notifications with zero operational editing access.

4. Comprehensive Audit Findings & Recommended Fixes

A rigorous scan across codebase architecture, API routes, security boundaries, and data synchronization identified the following items to address before production deployment:

ID	Severity	Component	Issue Description & Impact	Recommended Remediation
FIX-01	**MEDIUM**	Corridor Route *corridors.py*	Pydantic V2 class-based config deprecation warning emitted during test execution (`class Config: from_attributes = True`).	Migrate to Pydantic V2 modern `model_config = ConfigDict(from_attributes=True)` to prevent future deprecation breaks.

FIX-02	**MEDIUM**	Database Query *corridors.py*	N+1 query pattern in `list_corridors`: queries counts of assets, tasks, trains, and blocks separately per corridor in a Python loop.	Replace with single SQL `GROUP BY corridor_id` aggregation or joined subqueries to improve query response times.
FIX-03	**HIGH**	Concurrency & DB *session.py*	SQLite in development locks database file during simultaneous WebSocket writes and API task creations (WAL mode not enabled).	Enable SQLite WAL mode (`PRAGMA journal_mode=WAL;`) for local dev, and configure PostgreSQL connection string for production deployment.
FIX-04	**HIGH**	Auth & Security *AuthContext.tsx*	Role switching between Admin and Passenger operates strictly via local React state without backend JWT token signature checks.	Implement OAuth2/JWT bearer authentication on sensitive write routes (`/api/schedules/run`, `/api/tasks`, `/api/dispatch`).
FIX-05	**LOW**	WebSocket Feed *realtime.ts*	WebSocket reconnect logic uses fixed timer rather than exponential backoff with jitter.	Add exponential backoff (1s, 2s, 4s, max 16s) with random jitter to prevent server thundering herd on backend restarts.
FIX-06	**LOW**	Export Options *CompareView.tsx*	Schedule plan comparison displays metrics on screen but lacks 1-click export to official Indian Railways Divisional format (CSV / PDF).	Add direct CSV / PDF download action for Schedule Diff to facilitate divisional operating controller sign-off.

5. Verification Suite & Production Roadmap

The RailBlock AI platform has undergone thorough verification across frontend and backend modules:

Verification Check	Target Component	Status & Verification Proof
Backend Test Suite	FastAPI, OR-Tools, Validation (23 tests)	**PASSED (100%)** — All tests passed in 1.44s via Pytest 9.1.1
Frontend TypeScript Compilation	Vite, React 18, Tailwind, Recharts	**PASSED (0 Errors)** — `tsc -b && vite build` built in 2.36s
Pantone Theme Engine	CSS Tokens & Theme Provider	**VERIFIED** — Wheat (#F7F2E7) main, Magnetic Blue text, Carrot Orange accents, and Pure Black night mode
Sector Assignment Scalability	SectorAssignmentView.tsx	**VERIFIED** — Bounded scroll containers (max-h-580px), pagination controls, and Cards/Compact view toggle active
Admin Workflow Streamlining	DashboardView & Shell Navigation	**VERIFIED** — Redundant sidebar chips removed; grouped Operations/Planning/Admin navigation; 1-click Action Center enabled

Conclusion & Evaluation Summary:
RailBlock AI represents a mature, feature-complete research prototype for Indian Railways operations. With multi-departmental bundling across Engineering, TRD, and S&T;, an OR-Tools CP-SAT mathematical optimizer, and an intuitive, non-fatiguing Wheat-and-Carrot design system, the solution directly addresses SIH26027. Applying the recommended database WAL mode and JWT security fixes will transition the platform from prototype to operational field deployment.